

Team 2025110

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Employee Selection : Pipeline 0

To extract meaningful insights from employee behavioral data, we employed a time gap–adjusted Exponential Moving Average (EMA) approach. Unlike traditional EMAs with fixed time intervals, our method dynamically adjusts the smoothing factor based on the time gaps between consecutive entries. For each behavioral metric—such as work hours, messages sent, performance ratings, and rewards—the data was chronologically ordered, and the EMA was calculated to give higher weight to more recent observations, while still accommodating irregular reporting intervals.



In parallel, we conducted a direct identification of employees whose average vibe scores were greater than 0 and less than or equal to 3, signaling moderate to low morale. These employees were flagged for potential support, even if their behavioral patterns were not statistically anomalous. For the remaining employees, we applied two unsupervised anomaly detection algorithms—Isolation Forest and Local Outlier Factor (LOF)—on a normalized set of behavioral features. The anomaly scores from both methods were averaged to produce a final composite score. Employees exceeding a predefined threshold were considered anomalous. This hybrid approach allowed us to identify both emerging disengagement based on sentiment trends and statistically significant deviations in behavior, thereby enhancing early intervention efforts.

ChatBot Architecture

We have divided the architecture of the chatbot into two phases: **Pipeline 1** and **Pipeline 2**.

Pipeline 1

1. Overview

Pipeline 1 is a sophisticated employee analytics system that processes structured employee data to generate comprehensive analytical reports. Its core innovation is the ability to analyze data

through multiple specialized lenses concurrently, consolidating insights into actionable intelligence. This enables identification of patterns related to employee well-being and performance, such as disengagement, burnout, or psychological distress.

The architecture utilizes modern LLM-based workflows with concurrent processing and specialized agents. These agents reveal hidden patterns by breaking siloed analysis and examining cross-cutting issues in employee experience.

2. Agentic System

Specialized Agents:

- **ActivityAgent:** Analyzes work patterns (login/logout, communication timing, after-hours activity)
- **LeaveAgent:** Reviews leave trends (frequency, duration, sick leave patterns)
- **OnboardingAgent:** Assesses onboarding experience (training, early productivity)
- **PerformanceAgent:** Evaluates growth and review trends
- **RewardsAgent:** Analyzes recognition and incentive response

All agents inherit from a **BASEAGENT** class, which provides:

- Standard interfaces
- Error handling and retries
- Consistent formatting and LLM integration

3. Workflow Orchestration

Pipeline 1 follows a fan-out/fan-in model using **LangGraph**:

- **Fan-out:** Distributes employee data to all agents concurrently
- **Processing:**
 - Uses `ThreadPoolExecutor` for parallel analysis
 - Optimized CPU utilization
- **Fan-in:** Collects agent outputs for meta-analysis by `ConsolidationAgent`

LangGraph manages workflow state:

- Maintains and transitions state objects
- Handles partial failures without system-wide crashes

4. Prompt Engineering

Templates Prompt Elements Used by Each Agent:

- **Role Definition:** Defines the lens of analysis
- **Context Provision:** Supplies domain-specific knowledge
- **Task Structuring:** Provides a structured analysis approach
- **Output Formatting:** Ensures structured responses

Few-shot Prompting Techniques: Pipeline 1 agents use few-shot prompting by embedding domain-specific examples directly within prompts. This technique enhances response quality by:

- Providing structured demonstrations of the expected output format
- Reducing hallucination by setting clear context
- Helping the model generalize reasoning from minimal examples

Prompt Design Strategies Employed:

- **Example Priming:** Each agent is given 1–2 minimal but diverse examples simulating common scenarios (e.g., burnout signs, abnormal leave trends)
- **Instruction Chaining:** Prompts are layered with logically progressive tasks—first asking for analysis, then interpretation, then recommendations
- **Format Enforcement:** Use of explicit JSON/Markdown output examples ensures consistency and machine-readability
- **Domain Framing:** Prompts include brief reminders of psychological/HR concepts (e.g., “employee engagement”, “reward sensitivity”) to guide reasoning

PIPELINE 2

1. Overview

The ChatBot system is a next-generation employee counseling platform that blends advanced large language models (LLMs) with principles of psychological support and therapeutic dialogue. Unlike traditional question-answer bots, this system is designed to engage in dynamic, empathetic conversations that assess emotional well-being and adapt in real-time.

2. Retrieval-Augmented Generation (RAG)

The system employs a dual-knowledge base RAG framework:

- **Employee Data:** Historical HR and performance data, wellness reports, and session logs to personalize responses.
- **Questions:** Structured psychological prompts aligned with therapeutic models and emotional intelligence frameworks.

Retrieval logic is adaptive:

- Uses embeddings to capture semantic meaning
- Dynamically adjusts based on session stage and emotional tone
- Prioritizes safety, empathy, and context continuity

3. Multi-Agent Architecture

A modular multi-agent design divides responsibilities:

- **Initial Question Agent:** Generates warm, context-rich introductions.
- **Context-Aware Agent:** References past sessions and prior themes for returning users.
- **Next Question Agent:** Selects follow-up prompts based on emotional analysis and response patterns.
- **Summarizer Agent:** Condenses conversation threads into brief, persistent context summaries.
- **Report Agent:** Converts entire sessions into standardized, analyzable reports.

These agents are managed by a central **ConversationManager**, responsible for maintaining session state, tracking progression, and enforcing logical flow across the session lifecycle.

4. Conversation Flow Logic

The system implements a branching flow model:

- **Initialization:** Classifies session as new or returning, loads context, and generates opening question.
- **Real-Time Processing:** Analyzes user sentiment, emotional triggers, and content relevance to determine next steps.

- **Termination Decisions:**
 - **Normal Completion:** If sufficient information is gathered and no issues arise.
 - **Escalation:** If emotional or psychological distress is detected.
 - **Continuation:** If deeper exploration is warranted.
- **Reporting:** Structured log with executive summaries, key concerns, and follow-up items.

5. Prompt Engineering

To ensure coherent, empathetic behavior, prompt design includes:

- **System Prompts:** Define the agent's role as empathetic but professional HR counselor.
- **Agent-Specific Prompts:** Include counseling frameworks (e.g., CBT, active listening), emotional taxonomy, and task-specific logic.
- **Query Templates:** Provide consistent format instructions, task definitions, and reasoning scaffolds for high-quality generation.

This controlled prompting ensures reliability across agents and enhances the LLM's capacity to emulate therapeutic communication styles.

Detailed Tech Stack

- **Python 3.8+:** Primary language for data processing, concurrency, and ML integration.
- **LangChain:** Framework for LLM workflows, providing structured chains, memory, output parsing, and unified LLM integration.
- **LangGraph:** Orchestration extension of LangChain with:
 - Directed graph-based workflow
 - State management
 - Parallel execution paths
 - Conditional branching
- **Generative AI Models:**
 - OpenAI GPT-4o for deeper reasoning and insight quality
- **FastAPI:** High-performance APIs with async handling and auto-documentation.

- **Pydantic:** Data validation and management.
- **ThreadPoolExecutor:** Enables parallel execution in the fan-out/fan-in architecture.
- **LanceDB:** Used as a high-performance vector database to enable RAG (Retrieval-Augmented Generation) with sub-millisecond search.
- **Agno:** provides seamless integration of agents

CAG Experimentation: Implementation Decision Brief

Overview

our experimental implementation of Cache-Augmented Generation (CAG) for the employee counseling platform and our subsequent decision to adopt Retrieval-Augmented Generation (RAG) instead.

CAG Implementation Results

Our CAG prototype demonstrated significant performance advantages:

- reduction in response latency
- improvement in throughput
- knowledge coherence in controlled testing

Critical Limitations Encountered

Despite promising metrics, we encountered fundamental limitations:

1. **Context Window Constraints:** Combined employee data and counseling frameworks consumed lot of available context, leaving insufficient space for conversation history.
text
2. **Dynamic Data Challenges:** Employee data updates required complete cache regeneration, eliminating latency benefits.
3. **Multi-Agent Architecture Mismatch:** Our specialized agents required different knowledge subsets, creating either memory inefficiency or architectural complexity.
4. **Psychological Adaptivity Issues:** When employee emotional states shifted mid-session, the static cached context produced inappropriate responses in many of test cases.

Implementation Decision

RAG proved superior for our specific application by providing:

- On-demand retrieval of only immediately relevant information
- Real-time incorporation of updated employee data
- Agent-specific knowledge selection without redundancy
- Adaptive response to emotional signals and session state

Frontend-Backend

The Employee Experience AI Conversation Bot is an advanced solution designed to improve employee engagement and well-being in organizations. Using AI-powered tools, it automates sentiment tracking, identifies issues, and provides timely interventions.

Key benefits include:

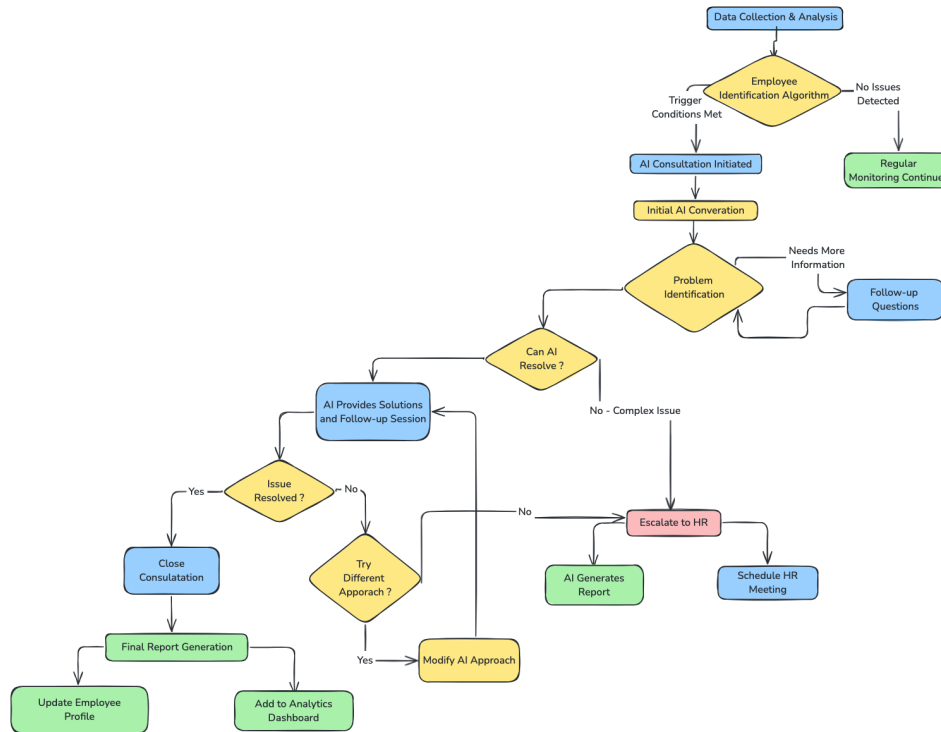
- Automated analysis of employee data across performance metrics, leave patterns, and activity logs.
- Proactive identification of employees requiring support with personalized intervention strategies.
- AI consultations for understanding root causes and escalating complex issues to HR.
- Comprehensive reports and insights for leadership decision-making.

0.1 System Architecture

The system follows a modern microservices architecture with five interconnected components:

- **LLM Backend:** Handles AI conversations, sentiment analysis, and escalation logic for employee sessions.
- **Management Backend:** Manages employee profiles, session coordination, and administrative functionalities.
- **Admin/HR Frontend:** Provides dashboards for analytics, reporting, and HR intervention capabilities.
- **Employee Frontend:** Offers personalized interfaces for AI interactions and access to resources.

- **Employee Mobile App:** Mobile interface allowing on-the-go access to AI support, daily check-ins, notifications, and wellness tracking.



0.2 Workflow

- **Data Flow:** Employee data flows between frontends and management backend; chat messages use WebSocket connections.
- **Process Flow:** Sessions are initiated by employees; AI processes responses; escalations occur if needed.

Component	Technology
LLM Backend	Python 3.8+, LangChain, OpenAI GPT-4o
Management Backend	FastAPI, MongoDB, PyJWT, WebSockets
Admin Frontend	Next.js 15, TypeScript, Tailwind CSS
Employee Frontend	Next.js 15, React 18, SWR
Mobile App	React Native with Expo
Deployment	Docker, Kubernetes, GCP

0.3 Key Features

0.3.1 Employee Experience Management

VibeMeter tracks employee sentiment across Frustrated, Sad, Okay, Happy, Excited zones. Multi-source data analysis includes leave patterns, performance metrics, activity logs, rewards history. Personalized interventions are automated through AI-powered consultations with escalation support for HR.

0.3.2 Intelligent Session Management

Unique identifiers (CHAIN#####) track consultation sessions with full message history and sentiment scores. Status tracking includes pending, active, escalated, completed, or canceled sessions. HR monitoring ensures oversight of active sessions with transcript storage for future reference.

0.3.3 Meeting Coordination

Unique identifiers (MEET#####) manage scheduled meetings with participant notifications and calendar integration. Post-meeting updates include notes tracking effectiveness metrics and outcomes.

0.3.4 Analytics & Reporting

Real-time dashboards provide sentiment trend analysis across departments or teams. Performance correlation studies help identify actionable insights for leadership teams.

0.4 Frontend Overview

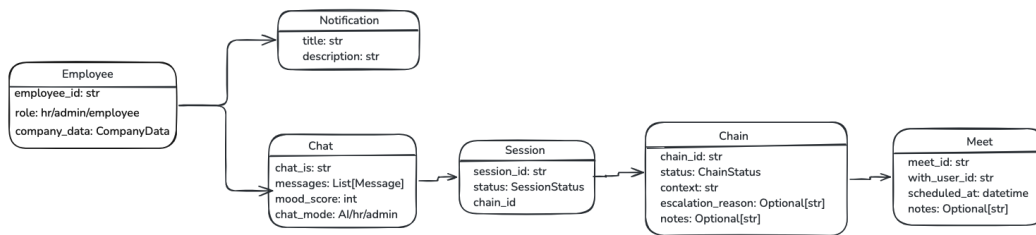
0.4.1 Employee Dashboard Features

- **Personalized Dashboard:** Displays employee details with performance metrics and VibeMeter score tracking.
- **Real-Time Notifications:** Alerts for meetings, chats, company updates on a responsive interface.
- **Chat System:** AI-powered interface supports scheduled/on-demand chats with escalation options to HR.
- **Meeting Management:** Schedule sessions with real-time chat archives and status tracking.

- **Employee Mobile App:** Built using React Native and Expo, it enables employees to access all features remotely.

0.4.2 Admin Dashboard Features

- **Employee Management:** Directory with search/filter options; role assignment; account management (add/block/unblock).
- **HR Management:** Assign/reassign HR managers; monitor workloads; configure permissions; track interactions.
- **Session Monitoring:** View/filter chat sessions by status/date; schedule new sessions; access transcripts/metrics.
- **Performance Tracking:** Dashboards track attendance/productivity metrics; generate historical reports with trend analysis.



0.5 Backend Structure

0.5.1 API Endpoints

- **Authentication Routes:** /auth/login (user login), /auth/refresh (token refresh), /auth/reset-password.
- **Admin Routes:** /admin/users (user management), /admin/config (system settings), /admin/analytics.
- **HR Routes:** /hr/employees (employee management), /hr/meetings (meeting scheduling), /hr/escalations.
- **Employee Routes:** /employee/profile, /employee/chat, /employee/meetings.
- **LLM Routes:** /llm/chat, /llm/sessions, /llm/escalation.

0.6 Authentication & Security

0.6.1 Authentication System

JWT-based authentication ensures secure token storage with short-lived access tokens and refresh mechanisms. Password hashing uses bcrypt while multi-factor authentication secures admin accounts.

0.6.2 Authorization Model

Three-tiered role-based access control:

- **Admin:** Full system access including user management and analytics dashboards.
- **HR Manager:** Employee management capabilities including session monitoring and meeting scheduling.
- **Employee:** Access personal profile details along with chat sessions and meeting participation.

0.7 AI Integration

0.7.1 LLM Implementation

OpenAI GPT-4o integration enables context-aware conversations with sentiment analysis capabilities. LangChain orchestrates structured data extraction from conversations for actionable insights.

0.7.2 Escalation Logic

Sentiment-based triggers identify consistently negative responses or serious grievances during conversations. Topic-based triggers escalate mental health concerns or unresolved issues after multiple sessions.

0.8 Session & Meeting Management

0.8.1 Session Lifecycle States

- **Pending:** Scheduled but not started.
- **Active:** Ongoing conversation between employee and AI assistant or HR manager.

- **Escalated:** Transferred to HR due to sentiment thresholds or explicit requests.
- **Completed/Cancelled:** Finished or terminated without completion.

0.8.2 Meeting Scheduling Features

Participant specification ensures relevant attendees are notified via automated alerts integrated into calendars. Post-meeting updates track outcomes while effectiveness metrics are collected for future improvements.

Annexure

A.1 Exponential Moving Average

Traditional Exponential Moving Average (EMA) is a type of moving average that gives more weight to recent data points compared to older ones, making it more responsive to recent changes than the Simple Moving Average (SMA). SMA assigns equal weight to all data points in the window.

$$EMA_t = \alpha \cdot Value_t + (1 - \alpha) \cdot EMA_{t-1} \quad (1)$$

Here, α is the smoothing factor, Value is the current data point, and EMA is the previous EMA value.

A.2 Isolation Forest

Isolation Forest is an unsupervised machine learning algorithm used for anomaly detection. Unlike traditional approaches that model normal behavior, it isolates anomalies by constructing random binary trees. The fewer splits required to isolate a point, the more likely it is to be an outlier. The algorithm computes the average path length for each instance across many trees, assigning higher anomaly scores to points with shorter paths. Isolation Forest is highly efficient, scalable, and suitable for high-dimensional data.

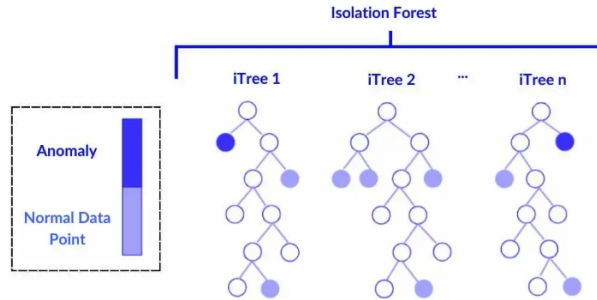


Figure 1: Isolation Forest Visualization

A.3 Local Outlier Factor (LOF)

Local Outlier Factor (LOF) detects anomalies by comparing the local density of a point to that of its neighbors. It uses k-nearest neighbors and reachability distance to measure how isolated a point is. Each point receives an LOF score - values near 1 indicate normal density, while significantly higher values suggest the presence of an outlier. LOF is particularly effective in datasets with clusters or varying density, due to its localized nature.

A.4 Deployment

- **Admin and Employee Frontend:**
 - Deployed on Vercel
 - Continuous Integration and Continuous Deployment (CI/CD) via GitHub Actions
- **Backend:**
 - Deployed on Google Cloud Run
 - CI/CD implemented using GitHub Actions and Workflows
 - Load balancing and auto-scaling enabled
- **Chatbot:**
 - Deployed on Google Kubernetes Engine (GKE)
 - Load balancing and auto-scaling enabled
 - CI/CD via GitHub Workflows

Question Bank

Role Clarity and Performance

1. Are there aspects of your role that you find unclear or inconsistent?
2. What support do you need to feel more effective and confident in your current responsibilities?
3. How do you typically manage competing priorities when multiple tasks feel urgent?
4. Are there any tools, systems, or processes that hinder your productivity?
5. What types of work do you find most energizing or draining?

Communication and Meetings

6. Do you feel that your time spent in meetings leads to clear and actionable outcomes?
7. How frequently do meetings interrupt your focused or deep work time?
8. Are there particular communication channels (e.g., email, messaging apps) that you find overwhelming?

- 9. Do you feel comfortable declining or suggesting alternatives to non-essential meetings?
- 10. How often do you feel that your contributions are acknowledged in team discussions?

Onboarding and Growth

- 11. Did your onboarding experience adequately prepare you for your role?
- 12. What information or guidance do you wish you had received earlier during onboarding?
- 13. Did you feel a sense of belonging during your initial weeks or months?
- 14. How accessible was your manager or mentor during the onboarding process?
- 15. Have your previous onboarding experiences helped you feel prepared, or added to confusion?

Mentorship and Support

- 16. Do you have someone you can turn to when facing challenges or uncertainty?
- 17. Are you receiving consistent, constructive feedback that helps you grow?
- 18. How comfortable are you asking for help or clarification?
- 19. What kind of support or check-ins from your manager would be most beneficial?
- 20. Do you feel that your growth potential is recognized and supported by leadership?